the `minikube ip --profile polar` command (as explained in chapter 9). On macOS, it's going to be `127.0.0.1`. Open a Terminal window, and run the following command (make sure you replace the `<ip-address>` placeholder with the cluster IP address, depending on your operating system):

```
$ echo "<ip-address> polar-keycloak" | sudo tee -a /etc/hosts
```

On Windows you must map the `polar-keycloak` hostname to `127.0.0.1` in the hosts file. Open a PowerShell window as an administrator, and run the following command:

```
$ Add-Content C:\Windows\System32\drivers\etc\hosts "127.0.0.1 polar-keycloak"
```

I have updated the scripts for deploying all the backing services for Polar Bookshop, including Keycloak and Polar UI. You can get them from the /Chapter11/11-end/polar-deployment/kubernetes/platform/development folder in the code repository accompanying the book (https://github.com/ThomasVitale/cloud-native-spring-in-action) and copy them into the same path in your `polar-deployment` repository. The deployment also includes the configuration of a dedicated Ingress for Keycloak, accepting requests directed to the `polar-keycloak` hostname.

At this point you can run the `./create-cluster.sh` script (polar-deployment/kubernetes/platform/development) to start a minikube cluster and deploy all the backing services for Polar Bookshop. If you're on Linux, you'll be able to access Keycloak directly. If you're on macOS or Windows, remember to run the `minikube tunnel --profile polar` command first. Either way, you can open a browser window and access Keycloak at polar-keycloak/ (include the final slash).

Finally, try running the entire system on Kubernetes after updating the deployment scripts for Edge Service to configure the URLs for Polar UI and Keycloak. You can refer to the Chapter11/11-end folder in the code repository accompanying the book to check the final result (https://github.com/ThomasVitale/cloud-native-spring-in-action).

The next chapter will expand on the subject of security. It will cover how to propagate the authentication context from Edge Service to the downstream applications, and how to configure authorization.

Summary